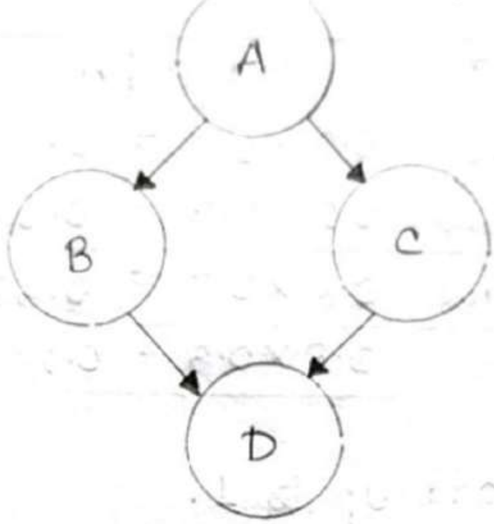
1. Consider the following Bayesian Network:

$P(A) = 0.75$



$P(B|A) = 0.2$
$P(B|\neg A) = 0.5$

$P(C|A) = 0.7$
$P(C|\neg A) = 0.25$

$$P(D | B \wedge C) = 0.3$$
$$P(D | B \wedge \neg C) = 0.25$$
$$P(D | \neg B \wedge C) = 0.1$$
$$P(D | \neg B \wedge \neg C) = 0.35$$

Here, A, B, C, D are Boolean random variables.
If we know that A is true, what is the probability of D being true?

Ans.: Since we are told that A is true, we only need the probabilities conditioned on A.

From the diagram,

$P(B|A) = 0.2$

$P(\neg B|A) = 1 - 0.2 = 0.8$

Also,

$P(C|A) = 0.7$

$P(\neg C|A) = 1 - 0.7 = 0.3$

Since B and C are conditionally independent from each other (given A), we can calculate their joint probabilities by multiplication.

B	C	P(B,C | A)
True	True	$0.2 \times 0.7 = 0.14$
True	False	$0.2 \times 0.3 = 0.06$
False	True	$0.8 \times 0.7 = 0.56$
False	False	$0.8 \times 0.3 = 0.24$

The results add up to 1.

$$0.14 + 0.06 + 0.56 + 0.24 = 1.00$$

Given probability table for D,

$$P(D \mid B, C) = 0.3$$
$$P(D \mid B, \neg C) = 0.25$$
$$P(D \mid \neg B, C) = 0.1$$
$$P(D \mid \neg B, \neg C) = 0.35$$

However, according to the question, we need : $P(D \mid A)$
Since we don't know whether B & C are true, we sum over all four possibilities:

$$\begin{aligned} P(D \mid A) = \; & P(D \mid B, C) \cdot P(B, C \mid A) \\ & + P(D \mid B, \neg C) \cdot P(B, \neg C \mid A) \\ & + P(D \mid \neg B, C) \cdot P(\neg B, C \mid A) \\ & + P(D \mid \neg B, \neg C) \cdot P(\neg B, \neg C \mid A) \end{aligned}$$

$$\Rightarrow P(D|A) = (0.3)\cdot(0.14) + (0.25)\cdot(0.06) + (0.1)\cdot(0.56) + (0.35)\cdot(0.24)$$

$$= 0.042 + 0.015 + 0.056 + 0.084$$

$$= 0.197$$

Therefore, if A is true, the probability of D being true is:

$$\boxed{P(D|A) = 0.197}$$

② Suppose a Genetic Algorithm uses chromosomes of the form $x = abcdefg$ with a fixed length of 8 genes. Each gene can be any digit between 0 to 9. Let the fitness of individual x be calculated as:

$$f(x) = (a+b) - (c+d) + (e+f) - (g+h)$$

Let the initial population consist of 4 individuals with the following chromosomes:

$x1 =$	6	5	4	1	3	5	3	2
$x2 =$	8	7	1	2	6	6	0	1
$x3 =$	2	3	9	2	1	2	8	5
$x4 =$	4	1	8	5	2	0	9	4

ⓐ Evaluate the fitness of each individual and arrange them in order with the fittest first and the least fit last.

Ans.: We are given chromosomes of the form,

$$x = abcdefgh$$

Fitness function,

$$f(n) = (a+b) - (c+d) + (e+f) - (g+h)$$

(i) <u>Individual x_1</u>

$x_1 =$

6	5	4	1	3	5	3	2
a	b	c	d	e	f	g	h

$$f(x_1) = (6+5) - (4+1) + (3+5) - (3+2)$$
$$= 11 - 5 + 8 - 5$$
$$= 9$$

(ii) <u>Individual x_2</u>

$x_2 =$

8	7	1	2	6	6	0	1
a	b	c	d	e	f	g	h

$$f(x_2) = (8+7) - (1+2) + (6+6) - (0+1)$$
$$= 15 - 3 + 12 - 1$$
$$= 23$$

(iii) Individual x_3

x_3 =

2	3	9	2	1	2	8	5
a	b	c	d	e	f	g	h

$$f(x_3) = (2+3)-(9+2)+(1+2)-(8+5)$$
$$= 5-11+3-13$$
$$= -16$$

(iv) Individual x_4

x_4 =

4	1	8	5	2	0	9	4
a	b	c	d	e	f	g	h

$$f(x_4) = (4+1)-(8+5)+(2+0)-(9+4)$$
$$= 5-13+2-13$$
$$= -19$$

So, $f(x_1) = 9$

$f(x_2) = 23$

$f(x_3) = -16$

$f(x_4) = -19$

Arranging in order of the fittest to the least fit,

$$23 > 9 > -16 > -19$$

$$\Rightarrow \boxed{x_2 > x_1 > x_3 > x_4}$$

(b) Perform the following crossover operations :

i) Cross the first two individuals using one-point crossover at the middle point.

In question 2(a), we established the fitness ranking and arranged the individuals in their fitness order :

$$x_2 > x_1 > x_3 > x_4$$

So, the first two [fittest] individuals for crossover are : x_2 and x_1.

Since each chromosome has 8 genes, the middle is between the 4th and 5th gene :

$$a\ b\ c\ d \mid e\ f\ g\ h$$

Which means,

$$x_2 = 8\ 7\ 1\ 2 \mid 6\ 6\ 0\ 1$$
$$x_1 = 6\ 5\ 4\ 1 \mid 3\ 5\ 3\ 2$$

$$O_1 = 8\ 7\ 1\ 2 \mid 3\ 5\ 3\ 2$$
$$O_2 = 6\ 5\ 4\ 1 \mid 6\ 6\ 0\ 1$$

ii) Cross the second and third fittest individual using a two-point crossover (points b and f)

From question 2(a), we determined the second and third fittest individuals were, x_1 and x_3.

Each chromosome has 8 genes:

a b c d e f g h

Performing crossover cuts at points 'b' and 'f':

a b | c d e f | g h

We know, for two-point crossover, we exchange the section between two crossover points:

$O_3 \rightarrow x_1 | x_3 | x_1$

$O_4 \rightarrow x_3 | x_1 | x_3$

Therefore,

$x_1 = 65 | 4135 | 32$

$x_3 = 23 | 9212 | 85$

$O_3 = 65 | 9212 | 32$

$O_4 = 23 | 4135 | 85$

iii) Uniform crossover of the first and third fittest individual.

The first and third fittest individual:

$x_2 = 8\ 7\ 1\ 2\ 6\ 6\ 0\ 1$

$x_3 = 2\ 3\ 9\ 2\ 1\ 2\ 8\ 5$

Uniform crossover means each gene alternates between the crossover individuals:

if genes are denoted as 'g',

$g_1, g_2, g_3 \dots$

Genes of x_2 : $x_2g_1, x_2g_2, x_2g_3, \dots$

Genes of x_3 : $x_3g_1, x_3g_2, x_3g_3, \dots$

$O_5 \rightarrow x_2g_1\ x_3g_2\ x_2g_3\ x_3g_4 \dots$ and so on.

$O_6 \rightarrow x_3g_1\ x_2g_2\ x_3g_3\ x_2g_4 \dots$ and so on.

Therefore,

$$
\begin{aligned}
x_2 &= 8\,7\,1\,2\,6\,6\,0\,1 \\
x_3 &= 2\,3\,9\,2\,1\,2\,8\,5 \\
\hline
O_5 &= 8\,3\,1\,2\,6\,2\,0\,5 \\
O_6 &= 2\,7\,9\,2\,1\,6\,8\,1
\end{aligned}
$$

(c) Suppose the new population consists of six offspring individuals received by the crossover operations in the previous question. Evaluate the fitness of the new population, showing all your workings. Has the overall fitness improved?

Ans.: The six offsprings obtained from the previous question:

$O_1 = 87123532$
$O_2 = 65416601$
$O_3 = 65921232$
$O_4 = 23413585$
$O_5 = 83126205$
$O_6 = 27921681$

The fitness function:

$$f(x) = (a+b)-(c+d)+(e+f)-(g+h)$$

i) $O_1 = 87123532$

$a=8, b=7, c=1, d=2, e=3, f=5, g=3, h=2$

$\therefore f(O_1) = (8+7)-(1+2)+(3+5)-(3+2)$
$= 15$

ii) $O_2 = 65416601$

$a=6, b=5, c=4, d=1, e=6, f=6, g=0, h=1$

$\therefore f(O_2) = (6+5)-(4+1)+(6+6)-(0+1)$
$= 17$

iii) $O_3 = 65921232$

$a=6, b=5, c=9, d=2, e=1, f=2, g=3, h=2$

$\therefore f(O_3) = (6+5)-(9+2)+(1+2)-(3+2)$

$= -2$

iv) $O_4 = 23413585$

$a=2, b=3, c=4, d=1, e=3, f=5, g=8, h=5$

$\therefore f(O_4) = (2+3)-(4+1)+(3+5)-(8+5)$

$= -5$

v) $O_5 = 83126205$

$a=8, b=3, c=1, d=2, e=6, f=2, g=0, h=5$

$\therefore f(O_5) = (8+3)-(1+2)+(6+2)-(0+5)$

$= 11$

vi) $O_6 = 27921681$

$a=2, b=7, c=9, d=2, e=1, f=6, g=8, h=1$

$\therefore f(O_6) = (2+7)-(9+2)+(1+6)-(8+1)$

$= -4$

Fitness ranking: $17 > 15 > 11 > -2 > -4 > -5$

$\Rightarrow O_2 > O_1 > O_5 > O_3 > O_6 > O_4$

$\therefore$ fittest offspring $\rightarrow O_2$ [$\because f(O_2) = 17$]

the least fit offspring $\rightarrow O_4$ [$\because f(O_4) = -5$]

From question 2(a),

Original population:

$f(x_1) = 9$

$f(x_2) = 23$

$f(x_3) = -16$

$f(x_4) = -19$

$\therefore$ Total fitness $= 9 + 23 - 16 - 19$

$= -3$

Since there were 4 individuals, the average fitness was:

$-3/4 = -0.75$

New population:

$f(O_1) = 15$

$f(O_2) = 17$

$f(O_3) = -2$

$f(O_4) = -5$

$f(O_5) = 11$

$f(O_6) = -4$

$\therefore$ Total fitness $= 15 + 17 + (-2) + (-5) + 11 + (-4)$

$= 32$

Since there are 6 offsprings, their average fitness is: $32/6 = 5.33$

Clearly, $5.33 > -0.75$

Therefore, the average fitness increased from -0.75 to 5.33. The improvement in average fitness is,

$$5.33 - (-0.75) = 6.08$$

The total fitness also increased from -3 to 32. It can be said that the overall fitness has improved. However,

The original population had a best individual with fitness 23 (x_2), whereas the best offspring has fitness 17 (O_2).

So although the average fitness of the population improved significantly, the best individual actually became less fit.

(3) A logistics company wants to optimize the order-picking route inside a warehouse using a Genetic Algorithm (GA). The warehouse contains six storage locations labeled A, B, C, D, E and F. The distance matrix between the locations is given below:

	A	B	C	D	E	F
A	0	8	12	15	10	9
B	8	0	7	11	14	13
C	12	7	0	9	8	10
D	15	11	9	0	6	5
E	10	14	8	6	0	7
F	9	13	10	5	7	0

The picker starts and ends at A. The initial population consists of the following four chromosomes:

Chromosome	Route
C1	A-B-C-D-E-F-A
C2	A-C-E-B-F-D-A
C3	A-D-F-E-C-B-A
C4	A-E-B-F-D-C-A

A single point crossover is performed after the third gene between the two chromosomes having the highest fitness. No mutation is applied.

(a) Calculate total distance and fitness value of each chromosome.

We know,

$$\text{Fitness} = \frac{1}{\text{Total Distance}}$$

i) C1: A → B → C → D → E → F → A

From the distance matrix,

A → B = 8
B → C = 7
C → D = 9
D → E = 6
E → F = 7
F → A = 9

$\therefore D(C1) = 8+7+9+6+7+9 = 46$

$\therefore f(C1) = 46^{-1} \approx 0.02174$

ii) C2: A → C → E → B → F → D → A

A → C = 12
C → E = 8
E → B = 14
B → F = 13
F → D = 5
D → A = 15

$\therefore D(C2) = 12+8+14+13+5+15 = 67$

$\therefore f(C2) = 67^{-1} \approx 0.01493$

iii) C3 : $A \to D \to F \to E \to C \to B \to A$

$A \to D = 15$

$D \to F = 5$

$F \to E = 7$

$E \to C = 8$

$C \to B = 7$

$B \to A = 8$

$\therefore D(C3) = 15+5+7+8+7+8 = 50$

$f(C3) = 50^{-1} = 0.02$

iv) C4 : $A \to E \to B \to F \to D \to C \to A$

$A \to E = 10$

$E \to B = 14$

$B \to F = 13$

$F \to D = 5$

$D \to C = 9$

$C \to A = 12$

$\therefore D(C4) = 10+14+13+5+9+12 = 63$

$f(C4) = 63^{-1} \approx 0.01587$

Since higher fitness is better, the ranking is :

$C1 > C3 > C4 > C2$

(b) Select the two fittest chromosomes for reproduction.

From question 3(a), we found that the two fittest Chromosomes were,

C1 (0.02174) and C3 (0.02)

Their routes are,

C1 = A-B-C-D-E-F-A

C3 = A-D-F-E-C-B-A

(c) Perform a single-point crossover after the third gene to generate two offsprings.

The crossover point is after the third gene.

Parent 1 (C1): $\boxed{A-B-C}$ | D-E-F-A

Parent 2 (C3): $\boxed{A-D-F}$ | E-C-B-A

The portions before the crossover point are kept, while the portions after the crossover point are exchanged.

Therefore,

Parent 1 (C1): A-B-C | D-E-F-A

Parent 2 (C2): A-D-F | E-C-B-A

Offspring 1 (O_1): A-B-C | E-C-B-A

⇒ A-B-C-E-C-B-A

Offspring 2 (O_2): A-D-F | D-E-F-A

⇒ A-D-F-D-E-F-A

(d) Determine which chromosome (parent or offspring) has the best fitness after crossover.

From question 3(a),

$f(C1) = 0.02174$

$f(C3) = 0.02$

From question 3(c),

$O_1 = A-B-C-E-C-B-A$

$A \to B = 8$

$B \to C = 7$

$C \to E = 8$

$E \to C = 8$

$C \to B = 7$

$B \to A = 8$

$\therefore D(O_1) = 8+7+8+8+7+8 = 46$

$\therefore f(O_1) = 46^{-1} \approx 0.02174$

$O_2 = A-D-F-D-E-F-A$

$A \to D = 15$

$D \to F = 5$

$F \to D = 5$

$D \to E = 6$

$E \to F = 7$

$F \to A = 9$

$\therefore D(O_2) = 15+5+5+6+7+9 = 47$

$\therefore f(O_2) = 47^{-1} \approx 0.02128$

Now,

Chromosome	Fitness
C1 (Parent 1)	0.02174
C3 (Parent 2)	0.02
O_1 (offspring1)	0.02174
O_2 (offspring2)	0.02128

Since higher fitness is better,

$$0.02174 > 0.02128 > 0.02$$

∴ The best chromosomes are : C1 and O1 with the fitness value of $\boxed{0.02174}$